

AA 212: Advanced Feedback Control Design

Sean Lin
DATE 04/2026

I added deep proofs to this cheatsheet. Key results or math properties are on my github.io site

State-Space General

For ease of reference, a general SS dynamics model and its dimensions are as follows.

$$\begin{aligned}\dot{x}_{n \times 1} &= A_{n \times n}x_{n \times 1} + B_{n \times m}u_{m \times 1} \\ y_{p \times 1} &= C_{p \times n}x_{n \times 1} + D_{p \times m}u_{m \times 1}\end{aligned}$$

Where n is the dimension of the state space. p is the dimension of the measurement space, and m is the dimension of the control space.

As a practical note, many classic control systems like temperature or voltage control still use PID controllers and are powered by transfer functions rather than SS control. However, an overwhelming amount of aerospace GNC is done in SS realization.

Here is the full classical state-space controller block diagram we use in this course.

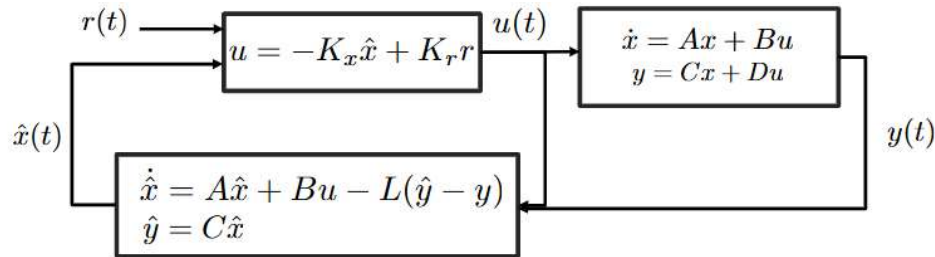


Figure 1: Full Classic State-Space Diagram

A quick legend of components inherent to the structure of the feedback loop.

1. A : the state dynamics matrix.
2. B : the control matrix.
3. C : the measurement matrix.
4. D : the pass-through matrix (usually $D = 0$).
5. x : the true state (**non-transparent**, but ultimately what we want to control)
6. $\hat{x}$: the state belief (transparent, what the system believes is the state).
7. y : the measurement the system receives (transparent, directly from sensors).
8. $\hat{y}$: the measurement the system expects (transparent, based deterministically on $\hat{x}$).

9. u : the control action.

10. r : the target signal (think of it as the guidance input in a GNC loop)

A legend of the components that are designed by the engineer.

1. K_x : The feedback gain(s). These are designed to place closed loop system poles where wanted.
2. K_r : The feed-forward gain(s). Note that these simply serve to scale the reference signal and so changing K_r will only ever change the magnitude of the response, not its transient or steady-state behavior.
3. L : The observer. Common observers are Luenberger, Kalman Filters (and their variants), or Particle Filters. These are designed to place closed loop observer poles where wanted. (think of the observer as the navigator in a GNC loop)

Similar State Spaces

A very useful insight moving forward is what happens if you decide you want to massage your states x into a new state $\tilde{x}$ via a LINEAR transformation (M is invertible).

$$x = M\tilde{x} \quad ; \quad \dot{x} = M\dot{\tilde{x}}$$

The resulting similar state-space realization is as follows.

$$\begin{aligned} \dot{\tilde{x}} &= \tilde{A}\tilde{x} + \tilde{B}u &\Rightarrow \dot{\tilde{x}} &= M^{-1}AM\tilde{x} + M^{-1}Bu \\ y &= \tilde{C}\tilde{x} + \tilde{D}u &\Rightarrow y &= CM\tilde{x} + Du \end{aligned}$$

The proof is very algebraically straightforward. To get started, simply substitute the first 2 equations into the original ss.

SS2TF and the MIMO case

The key result we are looking for here is

$$G(s)|_{p \times m} = C(sI - A)^{-1}B + D$$

We arrive here by taking the Laplace transform of our original realization and rearranging for $G(s) = \frac{Y(s)}{U(s)}$. The first step is outline below.

$$\begin{aligned} \mathcal{L}\{\dot{x}\} &= sX(s) = AX(s) + BU(s) \\ \mathcal{L}\{y\} &= Y(s) = CX(s) + DU(s) \end{aligned}$$

Notice that $G(s)$ is the same dimension as D . We are used to $G(s)$ being of the form $G(s) = k \frac{N(s)}{D(s)} \in \mathbb{R}$. When $G(s) \in \mathbb{R}_{p \times m}$, we are officially dealing with a MIMO system where

$$G(s) = \begin{bmatrix} g_{11}(s) & \cdots & g_{1m}(s) \\ \vdots & \ddots & \vdots \\ g_{p1}(s) & \cdots & g_{pm}(s) \end{bmatrix}$$

Where every single $g_{ij}(s)$ is an individual transfer function between output y_i and input u_j . There are 2 important claims proved in class for the MIMO transfer matrix $G(s)$.

- $D_{ij}(s) = D(s)$. Aka each transfer function in $G(s)$ shares the same denominator and hence the same poles
- The poles of every transfer function in $G(s)$ are $= \text{eig}(A)$

The proof is straightforward, leveraging the definition of the matrix inverse as $X^{-1} = \frac{\text{Adj}(X)}{\det(X)}$. Hence

$$C(sI - A)^{-1}B = \frac{C\text{Adj}(sI - A)B}{\det(sI - A)}$$

Where the numerator is a $p \times m$ matrix and the denominator is... voila!

Linear Algebra and Differential Equations

Similar Matrices

Two matrices A and $\tilde{A}$ are similar if there exists a matrix M such that

$$\tilde{A} = M^{-1}AM$$

Some notes.

- similarity is **ONLY** defined for square matrices.
- A and $\tilde{A}$ share the same eigenvalues, including the same Geometric and Algebraic multiplicities of the eigenvalues.
- A and $\tilde{A}$ have eigenvectors related by $v = M^{-1}\tilde{v}$
- A and $\tilde{A}$ share the same **Jordan Normal Form**

Note that intuitively, two matrices are similar if they perform the same transformation under different bases.

GM and AM of eigenvalues

Definition of Algebraic Multiplicity (AM): the number of times the eigenvalue is present in the characteristic polynomial (this is the form of eigenvalue multiplicity we are most familiar with)

Definition of Geometric Multiplicity (GM): the size of the nullspace of $(\lambda I - A)$. In other words, the number of independent eigenvectors associated with the repeated eigenvalue.

Some notes on the intuition and significance

- $GM \leq AM$ always
- Algebraic multiplicity can be coincidental, geometric multiplicity is more meaningful from a linear transformation perspective
- a matrix is **defective** if for any repeated eigenvalue λ , $GM < AM$.
- a defective matrix is **NOT diagonalizable**, but it can still be invertible.
- for the truly curious, a defective matrix that remains invertible is a “shear” matrix that only stretches in a subset of directions, but does not necessarily collapse a subspace. Think going from a rectangle to a parallelogram.

Diagonalizeability

Definition: A square matrix A is diagonalizable if it can be decomposed to

$$A = V\Lambda V^{-1}$$

Where

- Λ is a diagonal matrix of the eigenvalues of A

- V is a vertical eigenvector matrix (the columns are associated eigenvectors of A).
- V does NOT have to be normalized. any scaling done by V is undone by V^{-1}

Note that

- It is now clear why defective matrices ($\sum GM < \sum AM$) are not diag-able. V is not invertible for a reduced eigenvector space.
- Diag-ability is a property preserved under similarity transform. if A similar to B and A diag-able $\rightarrow B$ also diag-able.
- This general pattern is seen very often in lin-alg, and we are essentially applying A by applying its eigenvalues in a bases of its eigenvectors.

SVD

Linearization

This section will be brief.

Almost all true dynamics models will be nonlinear. In otherwords, they cannot be put into the form $\dot{x} = Ax + Bu$. The VERY common engineering workaround to this is to linearize.

If we are given a state that evolves according to

$$\dot{x} = \begin{bmatrix} f_1(x, u) \\ f_2(x, u) \\ \vdots \\ f_n(x, u) \end{bmatrix}$$

Where no assumptions are made on the linearity of the functions $f_i(x, u)$. We must **linearize about an equilibrium point**. This is because an linear approximation here will actually accurately reflect system behavior. The approach used is a first-order Taylor approximation.

$$A_{ij} = \left. \frac{\partial f_i}{\partial x_j} \right|_{x_{eq}, u_{eq}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}_{x_{eq}, u_{eq}}$$

$$B_{ij} = \left. \frac{\partial f_i}{\partial u_j} \right|_{x_{eq}, u_{eq}} = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \dots & \frac{\partial f_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \dots & \frac{\partial f_n}{\partial u_m} \end{bmatrix}_{x_{eq}, u_{eq}}$$

Note that linearization can also be applied to nonlinear measurement models where

$$y = \begin{bmatrix} g_1(x, u) \\ g_2(x, u) \\ \vdots \\ g_n(x, u) \end{bmatrix}$$

and $g_i(x, u)$ is also not assumed linear. The subsequent Jacobians for C and D are not explicitly written out.

Our linearized state-space model is defined as follows

$$\begin{aligned} \dot{x} &= A(x - x_{eq}) + B(u - u_{eq}) \\ (y - y_{eq}) &= C(x - x_{eq}) + D(u - u_{eq}) \end{aligned}$$

And if you redefine states such that $x - x_{eq} = \delta x = x$ (and similarly for u and y), the classic realization is reached.

A higher-order approximation of dynamic and measurement models than the first-order would involve the Hessian, **BUT this is highly undesirable, as we would depart from linear control theory and nothing we learn here applies anymore**. (Also, the Hessian for a vector state $x \in \mathbb{R}^n$ would be $n \times n \times n$ which nobody wants to deal with).

The Matrix Exponential and the SS solution

Just as the differential equation $\dot{x} = ax$ has the solution $x = x_0 e^{at}$, there is an analytical solution to the classic state-space realization $x(t)$ given x_0 and $u(t)$. It is as follows.

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau$$

This solution is never proved outright, but can be verified using properties of the matrix exponential and applying the Leibniz Integral rules (not covered in these notes).

There is beautiful structure to intuitively unpack in this though.

1. The first term is quite familiar, and indicates how x would have evolved under no control $u(t)$ (natural dynamics)
2. The INTEGRAND of the second term takes the effect of the control action on the state $Bu(\tau)$ and rolls its effect through the remaining time horizon by multiplying it by $e^{A(t-\tau)}$.
3. The effect of $Bu(\tau)$ is rolled out like this for ALL $0 < \tau < t$ and summed (forced response).

It is reassuring that C and D do not appear at all in this equation since it is a solution for how dynamics evolve. Also, let's unpack the matrix exponential.

The Matrix Exponential

The matrix exponential is simply a mathematical definition that follows the exact form of the infinite series for the scalar exponential. It is pointless to question the intuition behind what it means to take the exponent of a matrix.

$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k$$

Here are a few key reminders when interpreting this series

- $A^0 = I$ and $0! = 1$
- it is easy to convince yourself that this power series remains the same when you take its derivative with respect to A . never try memorizing this series knowing this.
- if dealing with e^{At} (the form we find it in the SS solution), substitute A^k with $A^k t^k$ in the series.
- $\frac{d}{dt} e^{At} = A e^{At}$. The proof of this is in pset 2 p1a and leverages a simple index-shift variable change $k' = k - 1$.

Cayley-Hamilton and the Jordan Normal Form

C-H Theorem statement

Theorem: for any **square** matrix A , A satisfies its own characteristic polynomial $p(s) = \det(sI - A) = 0$. In other words $p(A) = 0$

According to Erdos and by extension Prof. Schwager, this is one for the books.

The famous false proof is $\det(AI - A) = \det(0) = 0$ does not work simply because of dimension mismatch between a scalar determinant and a polynomial evaluated on a matrix which yields a matrix. Formalized, $\det(AI - A) = 0 \in \mathbf{R}$ and $p(A) \in \mathbf{R}^{n \times n}$

The Cayley Hamilton Theorem is directly provable for a diagonalizable matrix $A = V\Lambda V^{-1}$. One can convince themselves of this. The full proof is built up later with Jordan Normal Form.

C-H Polynomial Reduction

Theorem: For a **square** matrix A in an infinite order matrix polynomial $f(A)$, its order can be reduced to n to match A 's dimension.

$$f(A) = \sum_{k=0}^{\infty} c_k A^k \quad \rightarrow \quad f(A) = \sum_{k=0}^{n-1} \alpha_k(A) A^k$$

Where $\alpha_k(A)$ is a scalar coefficient that is a function of A .

Below is the structure of the proof of C-H polynomial reduction. It assumes the C-H theorem is correct to begin with. Repeat this loop recursively until order n achieved.

1. Leveraging C-H: $A^n = -(a_{n-1}A^{n-1} + \dots + a_1A + a_0I) = g_{n-1}(A)$ (convince yourself)
2. extract first term of infinite matrix power series $f(A) = c_N A^N + \sum_{k=0}^{N-1} c_k A^k$
3. Break down $A^N = A^{N-n} A^n$ and substitute the -1 order-reduced polynomial from step 1 we obtained via C-H into A^n . Notice we have argued $A^N = g_{n-1}(A) A^{N-n}$ which is order $N-1$
4. Bringing the extracted first term back into the sum will yield a COMPLETELY new set of c_k , where $c'_k = c_k + c_N a_{n-1-k}$ and for $a_{<0} = 1$.

The algorithm for finding the new coefficients is

Algorithm (Finding coefficients $\alpha_k(A)$):

$$\begin{bmatrix} f(\lambda_1) \\ \vdots \\ f(\lambda_n) \end{bmatrix} = \begin{bmatrix} \lambda_1^{n-1} & \lambda_1^{n-2} & \cdots & \lambda_1 & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ \lambda_n^{n-1} & \lambda_n^{n-2} & \cdots & \lambda_n & 1 \end{bmatrix} \begin{bmatrix} \alpha_{n-1} \\ \vdots \\ \alpha_0 \end{bmatrix}$$

$$f = \Phi \alpha$$

$$\alpha = \Phi^{-1} f$$

For k repeated roots, add $(k-1)$ rows each the derivative $\frac{d\Phi_i(\lambda_i)}{d\lambda_i}$ of the row above.

Some things to keep in mind for this algorithm

- Recall again that $f(A)$ is the original function of a matrix that we are approximating with an infinite power series.
- For systems where we apply $f(At)$, simply add a t multiplier to all elements of Φ EXCEPT for the last column, which remains $\lambda_i^0 t^0 = 1$.
- α is structured in reverse from f and Φ .
- A being diag-able is a requirement for this to work.

Jordan Normal Form

Recall the motivation of the Jordan Normal form is such that it is designed to handle the C-H Theorem for defective (non-diag-able) matrices.

Proof of full C-H Theorem

Controllability and Observability

Note that Controllability and Observability are both very strong concepts that are mathematically rigorous but for practical purposes a bit too rigid.

Definition of Controllability:

$$\forall t_1 < t_2 \quad , \quad \forall x_1, x_2 \quad | \quad \exists u(t) \quad : \quad x(t_1) = x_1, x(t_2) = x_2$$

Colloquially, we can reach any target state from any initial state in any arbitrary amount of nonzero time.

Definition of Observability:

$$\forall t_1 < t_2 \quad , \quad \forall u(t), y(t), t \in [t_1, t_2] \quad | \quad u(t), y(t), t_1, t_2 \rightarrow x(t_1)$$

Colloquially, if on any proper time interval, we are given the control input and measurement output, we can deduce the state at the start of the time interval. There is a fixation on the initial state $x(t_1)$, though theoretically speaking the entire state trajectory can be rolled out. However, practically speaking, rollouts will have noise.

Each of these concepts has a single rank test that can be performed. Both are well known and below.

$$\mathcal{C} = [B \quad AB \quad \cdots \quad A^{n-1}B] \quad ; \quad \mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

Here are some useful things to know about $\mathcal{O}$ and $\mathcal{C}$

- the system is controllable or observable if these matrices are full-rank.
- as a general linalg rule, row rank = column rank.
- $\mathcal{C}$ has the potential to be fat-rectangular for MIMO systems. In this case, $\mathcal{C}$ should be full **row rank** for the system to be controllable.
- $\mathcal{O}$ has the potential to be long-rectangular for MIMO systems. In this case, $\mathcal{O}$ should be full **col rank** for the system to be observable.

The proof of the validity of these matrices is long and explored in the subsequent sections because it heavily leverages the Cayley-Hamilton Theorem and the Gramian.

The Gramian

I have yet to have a solid grasp on the structure behind the Gramian, but its purpose is as a joining block in the proof of the Controllability and Observability rank tests.

Open-Loop $u^*(t)$

Recall that the state-space solution is

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau$$

Entertain the question of how to reach a state $x_2(t_2)$ from a starting state $x_1(t_1)$ (implied that the duration is $t_2 - t_1$). Kalman's equation for minimum-energy control postulates that

$$u^*(t) = B^T e^{A^T(t_2-t)}W_c^{-1}(t_1, t_2)\delta x_{12}$$

As-is this optimal control input begs the definition of both $W_c(t_1, t_2)$ and δx_{12} . We can substitute this expression back into the state-space solution to obtain

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}BB^T e^{A^T(t-\tau)}W_c^{-1}(t)\delta x_{12}d\tau$$

At this step, it is important to realize that $u^*(t)$ is **by design**, hence its currently undefined components are also **by design**. We design them as so.

$$\begin{aligned}\delta x_{12} &= x(t) - x_0 e^{At} \\ W_c(t) &= \int_0^t e^{A(t-\tau)}BB^T e^{A^T(t-\tau)}d\tau\end{aligned}$$

It is not difficult to verify that these definitions are valid for the above equation.

Definition of Gramians:

$$\begin{aligned}W_c(t) &= \int_0^t e^{A(t-\tau)}BB^T e^{A^T(t-\tau)}d\tau \\ W_o(t) &= \int_0^t e^{A^T(t-\tau)}C^T C e^{A(t-\tau)}d\tau\end{aligned}$$

Note that

- both are of size $n \times n$.
- $W_o(t)$ is the dual of $W_c(t)$, and vice versa.

Duality, CCF, and OCF

Duality

Duality is an important feature of SS representations that **allow us to design observers and controllers using the same procedure**. Formally, here are the key aspects of duality.

$$\begin{aligned}
 SS = \begin{bmatrix} A & B \\ C & D \end{bmatrix} &\Rightarrow SS_{dual} = \begin{bmatrix} A^T & C^T \\ B^T & D^T \end{bmatrix} = \begin{bmatrix} A_{dual} & B_{dual} \\ C_{dual} & D_{dual} \end{bmatrix} \\
 \mathcal{C} &= \mathcal{O}_{dual}^T \\
 \mathcal{O} &= \mathcal{C}_{dual}^T \\
 (SS_{dual})_{dual} &= SS \\
 TF(SS)^T &= TF(SS_{dual}) \\
 W_{o \text{ dual}}(t) &= W_c(t)^T = W_c(t) \\
 W_{c \text{ dual}}(t) &= W_o(t)^T = W_o(t)
 \end{aligned}$$

Let's add a few verbal pieces of intuition

1. A system is Controllable if its dual is Observable
2. A system is Observable if its dual is Controllable
3. For SISO systems, the transfer function is scalar-valued, hence the dual transfer function matches the original transfer function.

CCF

Controllable Canonical Form (CCF) is a useful way to massage a state-space representation into a form where transfer function polynomial coefficients are directly evident from the matrix values. This is critical for pole-placement of the closed-loop system.

Def (Control Canonical Form): We say a state space system is in control canonical form if it has the form:

$$A_{ccf} = \begin{bmatrix} -a_{n-1} & -a_{n-2} & \cdots & -a_1 & -a_0 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix} \quad B_{ccf} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$C_{ccf} = [b_{n-1} \quad b_{n-2} \quad \cdots \quad b_1 \quad b_0] \quad D_{ccf} = 0$$

where $(a_{n-1}, a_{n-2}, \dots, a_0)$ and $(b_{n-1}, b_{n-2}, \dots, b_0)$ are coefficients of the equivalent transfer function:

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \cdots + a_1s + a_0}$$

There is a 2nd common form of CCF where B_{CCF} and C_{CCF} are flipped via the exchange matrix M . A_{CCF} is flipped both top-bottom and left-right under this transformation.

$$A'_{CCF} = M^{-1}A_{CCF}M \quad ; \quad B'_{CCF} = M^{-1}B_{CCF} \quad ; \quad C'_{CCF} = C_{CCF}M$$

This new form requires no further elaboration other than saying it is the form if you reverse the order of your state x .

The natural follow up to CCF is asking how to arrive at it from any general ss-realization. How do you achieve $(A, B, C, D) \rightarrow (A, B, C, D)_{CCF}$? The similarity transform is as follows.

Theorem (CCF Transform): For any (A,B) controllable with $D = 0$:

$$A_{ccf} = P^{-1}AP \quad B_{ccf} = P^{-1}B$$

$$C_{ccf} = CP \quad D_{ccf} = 0$$

where

$$P = \mathbb{C} \begin{bmatrix} 1 & a_{n-1} & a_{n-2} & \cdots & a_1 \\ 0 & 1 & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & \ddots & a_{n-2} \\ \vdots & \vdots & \ddots & \ddots & a_{n-1} \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

OCF

We naturally follow CCF up with its dual OCF.

Def (Observer Canonical Form): We say a state space system is in observer canonical Form if it has the form:

$$A_{ocf} = \begin{bmatrix} -a_{n-1} & 1 & 0 & \cdots & 0 \\ -a_{n-2} & 0 & 1 & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ -a_1 & 0 & \cdots & 0 & 1 \\ -a_0 & 0 & \cdots & 0 & 0 \end{bmatrix} \quad B_{ocf} = \begin{bmatrix} b_{n-1} \\ b_{n-2} \\ \vdots \\ b_1 \\ b_0 \end{bmatrix}$$

$$C_{ocf} = [1 \ 0 \ \cdots \ 0 \ 0] \quad D_{ocf} = 0$$

where $(a_{n-1}, a_{n-2}, \dots, a_0)$ and $(b_{n-1}, b_{n-2}, \dots, b_0)$ are coefficients of the equivalent transfer function:

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \cdots + a_1s + a_0}$$

And its similarity transform

Theorem (OCF Transform): For any (A,C) observable with $D = 0$:

$$A_{ocf} = P_o^{-1}AP_o \quad B_{ocf} = P_o^{-1}B$$

$$C_{ocf} = CP_o \quad D_{ocf} = 0$$

where

$$P_o^{-1} = \begin{bmatrix} 1 & a_{n-1} & a_{n-2} & \cdots & a_1 \\ 0 & 1 & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & \ddots & a_{n-2} \\ \vdots & \vdots & \ddots & \ddots & a_{n-1} \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}^T \quad \mathbb{O} = \mathbb{O}_{ocf}^{-1}\mathbb{O}$$

PPC, PPO, and the Luenberger Observer